Data Literacy Among Charles University PhD Students: Are They Prepared for Their Research Careers?

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Abstract. The goal of our study, based on an extensive survey, is to discover the attitudes toward data sharing and research data management among Charles University doctoral students. The research was carried out as a part of a Data Literacy Multinational Study cooperation project. We have used a Czech version of the data literacy questionnaire presented by Gobinda Chowdhury et al. at the ECIL conference 2016. Results show that doctoral students from all disciplines are willing to share data, but they are not aware of open access principles. Doctoral students have confirmed they share their data at least among their research teams. The results do not show a significant difference in this practice between disciplines. Doctoral students prefer their research data to be preserved for further research.

Keywords: Information literacy · Data literacy · Data sharing
Data sharing practices

1 Introduction

In recent years, the availability and volume of scientific data have grown exponentially. Research funding bodies encourage, or even require, open publication not only of research results, but also publication of involved data as well. Data sharing is an essential component of the direction in which science is moving [1]. This situation creates a need for training in data literacy: a set of competencies that can be defined as the component of information literacy, which enables individuals to access, interpret, critically assess, manage, handle and ethically use data [2].

Before designing any instructional intervention, it is nevertheless necessary to assess the current level of data management awareness, competencies and skills of individual researchers. This is the group, on whom the intervention is focused. The situation in the Czech Republic in the field of scientific data and data literacy has not been assessed so far. Our paper's goal, based on an extensive survey, is therefore to discover the attitudes among doctoral students in the Czech Republic toward data sharing and management. We will use a sample from Charles University in Prague.

Charles University is the largest multidisciplinary, research-oriented institution of higher education in the Czech Republic. It consists of seventeen faculties (individual

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schools) and has more than 50,000 students, 7,500 of whom are doctoral students. The university has approximately 4,600 academics and researchers [3].

The issues of data storage and data access are quite new in the Czech Republic. Charles University does not have its own open research data repository. The university has only collaborated on building a technological infrastructure for data archiving and sharing within research teams in private repositories (ownCloud service) [4].

2 Literature Review

The body of literature on data literacy has been growing steadily for several years. Its core focuses on the definition of data literacy and its main competencies and required skills. As Koltay points out in his reviews of the data literacy field [5, 6], several other labels have been used throughout the literature for the data literacy concept; namely, data information literacy [7], science data literacy [8], data management literacy [9] and research data literacy [10]. There seems to be no definite agreement on the preferred terminology, but data literacy and data information literacy are the most commonly used terms.

The understanding and scope of data literacy has evolved over time; together with its relation to statistical literacy and information literacy. It was used at first only in relation to numerical data; meaning the same thing as statistical literacy or quantitative literacy [11] or even understood as part of statistical literacy [12]. Its meaning was later broadened to cover the collection, processing, manipulation, evaluation and usage of all kinds of scientific data [8]. The foundations of the current scope of data literacy were laid by Carlson et al. with their data information literacy that joined the researcher-as-producer aspect together with the until-then-used aspect of researcher-as-consumer of data. Their concept also changed the view of the data-statistical-information literacy triangle. In Carlson's approach, data information literacy builds on and reintegrates data, statistical and information literacy into an emerging set of skills [7]. The scope of the concept was accepted by other researchers, but the term data information literacy is used only by part of them (for example, see Refs. [13, 14]), while others stick to the term data literacy [2, 6] because of it being more clear and straightforward [6].

So far, the only research in the field of research data in the Czech Republic that has some implications for the data literacy is the dissertation by Pavlaskova [15]. Her work focused primarily on research data from the viewpoint of long-term preservation and archiving, but in the process of her research, she examined different types of data by analysing data appendices from Charles University (Prague) dissertations.

3 Methods

The research was carried out as part of a Data Literacy Multinational Study cooperation project. We used a Czech version of the data literacy questionnaire presented by Chowdhury et al. [16] at the ECIL conference 2016. We distributed it by email to all academics and PhD students affiliated with Charles University.

The structure of the Charles University in Prague is a little different from the institutions the questionnaire has been intended for. Charles University consist of many various faculties, which offer doctoral studies in many disciplines. PhD students were sometimes not able to select his/her discipline listed in the questionnaire. However, the questionnaire is a very suitable starting point for more in-depth research that will follow, and it also offers the possibility of cross-country comparison on a scale till now unparalleled. The questionnaire considers researchers as both producers and consumers of data and focuses on three main topics:

- production and use of data (types of data and volume of data);
- attitudes toward data sharing and data sharing behaviour;
- awareness of various aspects of data sharing (metadata, identifiers and versioning) and data sharing practices.

Data, in the context of this study, is understood very broadly (in line with the definition given by the National Academy of Sciences) as facts, numbers, letters and symbols that describe an object, idea, condition, situation or other factors (National Academy of Sciences, 1999) and independent of its form. It reflects the practices of both natural sciences on one hand and humanities and social sciences on the other.

4 Survey Sample

We decided to use the "census" polling method because of the heterogeneity of the target group. We used the email addresses of academics and doctoral students taken from the Charles University information system.

We collected 1,434 completed questionnaires, 826 of which came from doctoral students and formed the basis for the following analysis (it is approximately 11% of all PhD students at the Charles University).

After we cleaned the data, we always separated the results into for basic scientific areas: humanities (including history and philology), medicine, natural sciences (including mathematics, physics and biology) and social sciences.

5 Research Questions and Hypotheses

As data management is an integral part of science nowadays, we presumed that the willingness and ability to share data and archive them correctly is one of the prerequisites for a successful research career. Hypotheses formulation is therefore guided by a general point of view to consider the current attitudes of doctoral students toward research data sharing and toward descriptions of current practices.

- H1: CU doctoral students are willing to share research data and support open access to data.
- H2: Research teams currently share data, but most probably there exist fundamental differences between subjects.
- H3: Doctoral students view research data archiving as an important prerequisite to further research work.

We used a five-point scale for several questions in the questionnaire. However, for clearer presentation, we merged the answers "I strongly agree" and "I agree" as YES and "I disagree" and "I strongly disagree" as NO.

6 Results Analysis

6.1 CU Doctoral Students are Willing to Share Research Data and Support Open Access to Data (H1)

The research into the attitudes toward research data sharing and the willingness of doctoral students to share data and toward open access produces interesting results. Results show that 64% (n = 826) of doctoral students are not familiar with open access issues. Nevertheless, as further results show, doctoral students do not lack an openness and a willingness to share data (Fig. 1).

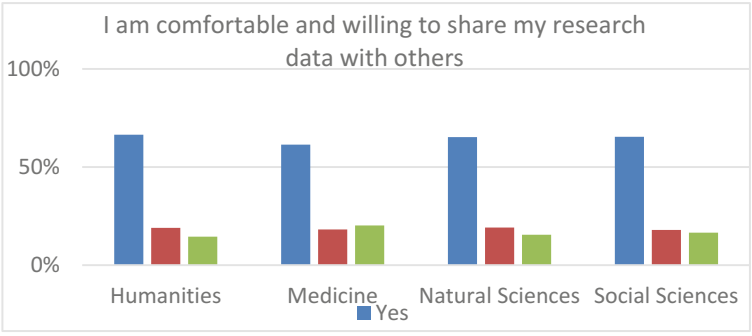


Fig. 1. Willingness to share data

The next attitudinal question on research data sharing concerned doctoral students' opinions on the issue of potential ethical problems that can arise in the process of data sharing, such as whether open data sharing can be contrary to ethical principles (for example, personal data protection). Doctoral students perceive this problem unequivocally as acute, and they mostly expressed that concern (Fig. 2).

As for other potential problems that can arise regarding data sharing, most respondents fear misinterpretation (37%) and misuse of data (33%). On the other hand, lack of policy and lack of resources do not cause concern.

6.2 Research Teams Currently Share Data, But Most Probably There Exist Fundamental Differences Between Disciplines

Outside its research into general attitudes, the questionnaire explored current data sharing practices as well. Here, we see the difference between the humanities and social sciences on one hand, where 36% (n = 179 for humanities, n = 139 for social sciences) of doctoral students do not share data, and medicine and the natural sciences on the other, where only five percent (n = 148) and ten percent (n = 360) report not sharing

data. However, the other answers show that most doctoral students from all disciplines share data in some way (Fig. 3).

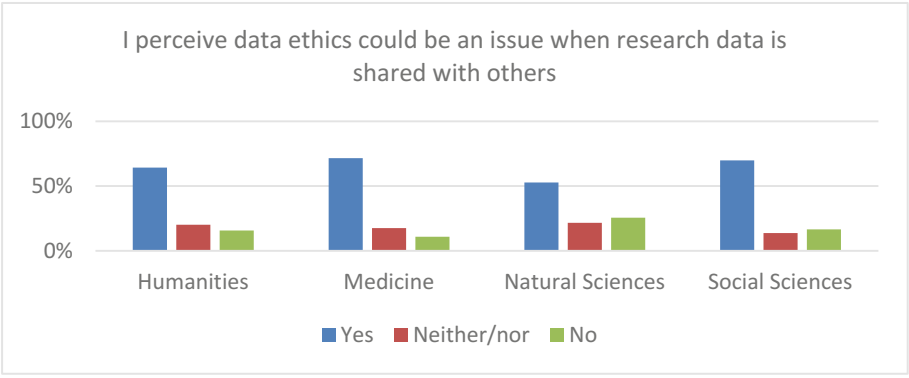


Fig. 2. Data sharing and ethical issues

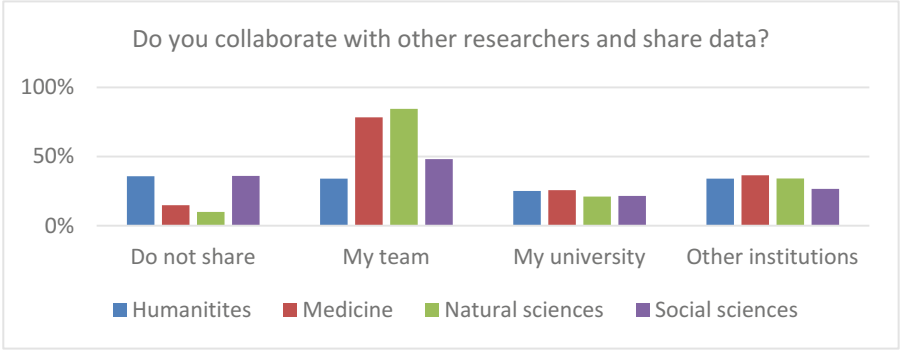


Fig. 3. Data sharing practices

Sharing data does not necessarily mean open access to data, as is shown by answers to the next question on data sharing. The most frequent means of sharing data is upon request. On the contrary, open access is the least frequent method. The smallest share was observed in medicine, with only two percent (n = 148) reporting use of the latter practice. Given the willingness of doctoral students to share data, this situation is more likely caused by a lack of easy solutions for data sharing or ignorance of them (Fig. 4).

6.3 Doctoral Students View Research Data Archiving as an Important Prerequisite to Further Research Work (H3)

On the issue of data archiving, even after a project wherein data was collected or created had concluded, doctoral students expressed themselves relatively unequivocally. The majority of students would welcome the possibility of long-term data

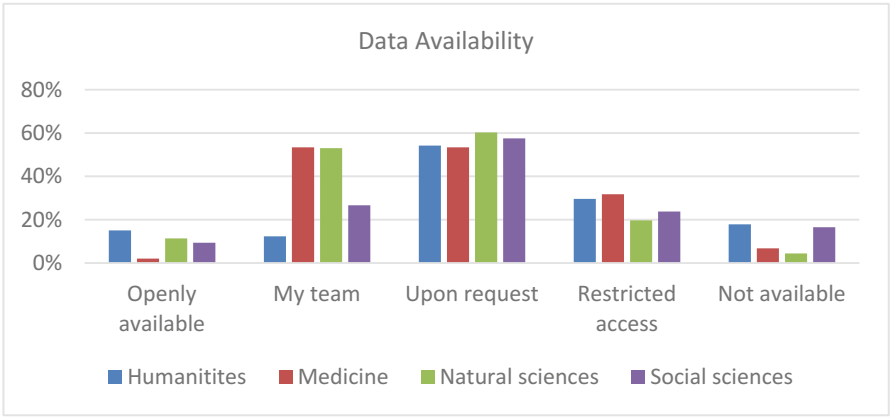


Fig. 4. Data availability

archiving. This question is of the utmost importance to the university. It points to the fact that doctoral students would appreciate the existence of a university data repository. Overall, 90% (n = 826) of all respondents want to archive their data (Fig. 5).

Answers to the question on current data storage practices show that an absolute majority of doctoral students store their data on their own devices (PCs and tablets). The differences between subject areas are negligible. 35–45% of doctoral students use external repositories; usage of cloud-based storage is on the same level. Only in social sciences does use of cloud-based storage exceed 40% (namely 46%, n = 139). Central repository usage appeared in some answers, although the university does not have any central repository as we have already mentioned (Fig. 6). Respondents probably regard subject repositories or repositories of their institutes as central.

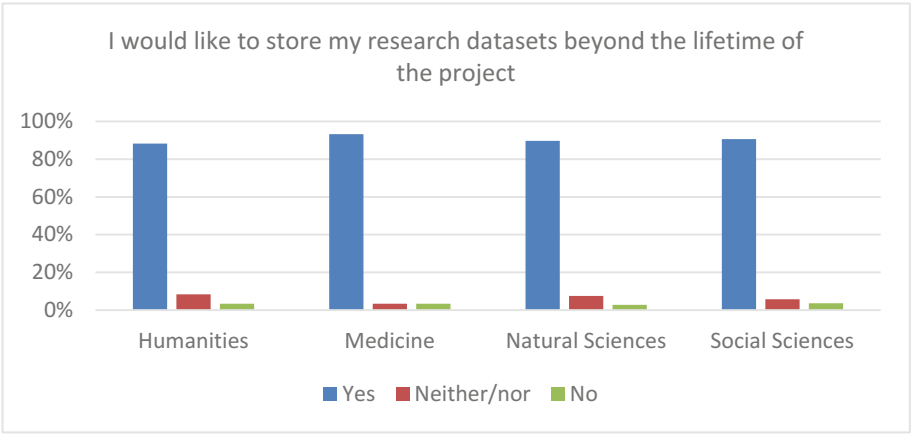


Fig. 5. Research data preservation

Even if they want to store their data, the doctoral students do not in most cases use metadata for data description in their current data archiving practices. Although 60% (n = 826) declare, that they know the term metadata, 39% do not use them at all and 34% (n = 826) rarely.

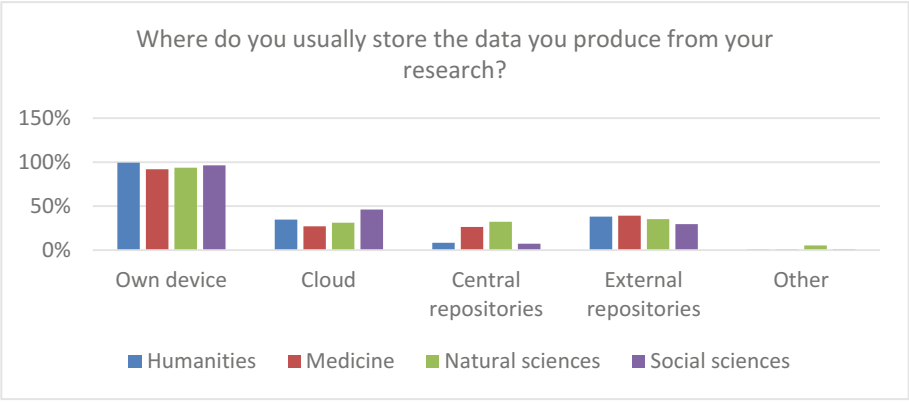


Fig. 6. Data storage

The volume of data that should be stored is substantial for the future building of a central university repository. Currently, most doctoral students use data amounting to numbers in mega or gigabytes. The same volume applies to the data resulting from their research.

7 Limitations of the Study

As the selection may be biased in favour of individuals who already have an interest in data literacy and who may consider data literacy skills more relevant to their work than individuals who do not have such interests, the results cannot be considered a general view in relation to the research population of the Czech Republic.

8 Discussion

We presumed that CU doctoral students are willing to share research data and support open access to data. This hypothesis has been confirmed only partially. The most important finding is, that doctoral students from all disciplines are willing to share data and that they already share them in some way. The concerns expressed in the connection with data sharing, mainly data misuse and misinterpretation and ethical issues, correspond to findings of other studies, for example Tenopir [1]. The difficulty is, that doctoral students in our study are not aware of open access principles and so the

requirements for open access to data, that are part of some funding schemes, might cause problems to them.

We also assumed that research teams currently share data, but there likely exist fundamental differences between subjects. Doctoral students have confirmed they share their data at least among their research teams, most frequently upon request. Nevertheless, the results do not show a significant difference in this practice between disciplines. There are probably other factors that influence the willingness to share data, as is shown for example in a study by Kim and Stanton [17].

Our last hypothesis was that doctoral students view research data archiving as an important prerequisite to further research work. This hypothesis has been confirmed. Doctoral students prefer their research data to be preserved for further research. However, they are not skilled in data management, which is probably a consequence of the fragmented approach to data management and data literacy issues at the Charles University.

9 Conclusion

At Charles University, doctoral students are willing to share data. Their attitudes show that they understand this task as an important one in their research careers. They would prefer data preservation for further research to take place in a common university data repository. The results show, on the other hand, that doctoral students are not aware of new trends in data sharing. First of all, they are unfamiliar with open access principles and principles concerning data management. We can say, that they are prepared only partially for their research careers, as they understand the importance of data in their research and they are prepared to share them, but they lack practical data management skills needed to manage their data effectively. For their better preparedness, it will be necessary to introduce data management skills training for doctoral students at the university.

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